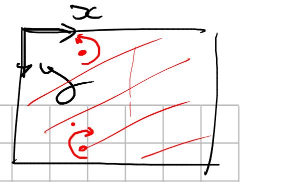
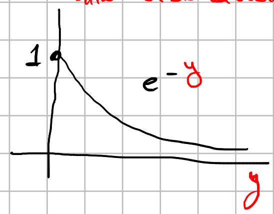
$$\underline{u} = \begin{bmatrix} u(x,y,\cancel{z},t) \\ v(x,y,\cancel{z},t) \\ w(x,y,\cancel{z},t) \end{bmatrix}$$



dirección de la onda $x+$

$$\begin{cases} u = A\left(e^{\alpha k y} + \gamma \cdot e^{\beta k y}\right) \cdot \cos\left[k\left(x - c_R t\right)\right] \\ v = A\left(\alpha e^{\alpha k y} + \xi e^{\beta k y}\right) \cdot \cos\left[k\left(x - c_R t\right)\right] \\ w = 0 \end{cases}$$



$\alpha = -0{,}8475$

$\beta = -0{,}39333$

$\gamma = -0{,}5773$

$\xi = 1{,}4679$

$\beta \cdot \gamma + \xi + 2\alpha = 0$

$\beta \xi - \gamma = 0 \Rightarrow \gamma = \beta \cdot \xi$

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

$$\begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} \\ \varepsilon_{xy} & \varepsilon_{yy} \end{bmatrix} \to 0$$

$$\varepsilon_{xx} = \frac{\partial u_x}{\partial x} \to \frac{\partial u}{\partial x} = A\left(e^{\alpha k y} + \gamma \cdot e^{\beta k y}\right) \cdot -\operatorname{sen}\left(k\left(x - c_R t\right)\right) \cdot k$$

$$= -Ak\left(e^{\overset{\alpha<0}{\alpha k y}} + \gamma \cdot e^{\overset{\beta<0}{\beta k y}}\right) \cdot \operatorname{sen}\left(k\left(x - c_R t\right)\right)$$

$$\varepsilon_{yy} = \frac{\partial u_y}{\partial y} = \frac{\partial v}{\partial y} = Ak \cdot \left(e^{ky\alpha} \cdot \alpha^2 + e^{ky\beta} \cdot \beta \cdot \xi\right) \cdot \cos\left(k\left(x - c_R t\right)\right)$$

$$\varepsilon_{xy} = \frac{1}{2}\left(\frac{\partial \overset{u}{u_x}}{\partial y} + \frac{\partial \overset{v}{u_y}}{\partial x}\right) = A \cdot \left(e^{ky\alpha} - e^{ky\beta}\right) \cdot k\alpha \cdot \cos\left(k\left(x - t c_R\right)\right)$$

1) Deformaciones ppales

$\left.\begin{matrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{matrix}\right\} \to \varepsilon_{máx}(x,y) \to$

$$\frac{\partial \varepsilon_{máx}}{\partial x} = 0 \qquad \frac{\partial \varepsilon_{máx}}{\partial t} = 0$$

$$\frac{\partial \varepsilon_{min}}{\partial y} = 0$$

$$\begin{matrix} \overset{\varepsilon}{\sigma}_{max} \\ \overset{\varepsilon}{\sigma}_{min} \end{matrix} = \frac{\overset{\varepsilon_x}{\sigma_x} + \overset{\varepsilon_y}{\sigma_y}}{2} \pm \sqrt{\left(\frac{\overset{\varepsilon_x}{\sigma_x} - \overset{\varepsilon_y}{\sigma_y}}{2}\right)^2 + \overset{\varepsilon_{xy}}{\tau_{xy}^2}}.$$

2) $\left.\begin{matrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{matrix}\right\}$

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta,$$

$$\to \begin{cases} \dfrac{\partial \varepsilon_{x'x'}}{\partial \theta} = 0 \\ \dfrac{\partial \varepsilon_{x'x'}(x,y,t)}{\partial x} = 0 \\ \dfrac{\partial \varepsilon_{x'x'}(x,y,t)}{\partial y} = 0 \end{cases}$$

$$\frac{\partial \varepsilon_{x'x'}}{\partial t}(x,y,t) = 0$$

En $y=0 \Rightarrow \varepsilon_{xx}, \varepsilon_{yy}$ son las deformaciones principales

$y=0$.

$$\varepsilon_{xy} = A.(\overbrace{e^{K.0.\alpha}}^{1} - \overbrace{e^{K0\beta}}^{1}) .K.\alpha.\cos[K(x-t.C_R)] = 0 \Rightarrow$$

$$\varepsilon_{xx} = -AK(1+\gamma).\overbrace{\operatorname{sen}[K(x-tC_R)]}^{\pm 1}$$

$$\varepsilon_{yy} = AK(\alpha^2+\gamma).\underbrace{\operatorname{sen}[K(x-tC_R)]}_{\pm 1}$$

$$\varepsilon_{xx} = \mp AK(1+\gamma) = \mp 0,4227\,AK$$

$$\varepsilon_{yy} = \pm AK(\alpha^2+\gamma) = \pm 0,1410\,AK$$

en distintos instantes los valores extremos

$\varepsilon_{min} = -0,4227\,AK$, $\varepsilon_{max} = 0,1410\,AK$

$\varepsilon_{min} = -0,1410\,AK$, $\varepsilon_{max} = 0,4227\,AK$

$A = 0,4$
$K = 2$
$C_R = 1$